

# محمد عابر زكريا عوض (SW)

---

```
using System;
using System.Linq;

class Stats
{
    static void Main()
    {
        int[] d =
{115,182,191,31,196,1099,5,172,10,179,83,21,20,21,186,177,195,193,188,199,62
,109,102};

        Array.Sort(d);
        int n = d.Length;

        double mean = d.Average();
        double median = n % 2 == 0 ? (d[n / 2] + d[n / 2 - 1]) / 2.0 : d[n /
2];

        double var = d.Select(x => Math.Pow(x - mean, 2)).Average();
        double std = Math.Sqrt(var);
        double q1 = Med(d.Take(n / 2).ToArray());
        double q3 = Med(d.Skip((n + 1) / 2).ToArray());
        double iqr = q3 - q1;

        Console.WriteLine($"Mean={mean}, Median={median}");
        Console.WriteLine($"Var={var}, Std={std}, Q1={q1}, Q2={median},
Q3={q3}");
        Console.WriteLine($"P20={Per(d, 20)}, P50={Per(d, 50)},
Range={d.Max() - d.Min()}, IQR={iqr}");
    }

    static double Med(int[] a) => a.Length % 2 == 0 ? (a[a.Length / 2] +
a[a.Length / 2 - 1]) / 2.0 : a[a.Length / 2];

    static double Per(int[] a, int p)
    {
        double i = (p / 100.0) * (a.Length - 1);
        int l = (int)Math.Floor(i), u = (int)Math.Ceiling(i);
        return l == u ? a[l] : a[l] + (i - l) * (a[u] - a[l]);
    }
}
```

```
}
```

```
using System;
using System.Linq;

class Outliers
{
    static void Main()
    {
        int[] d =
{115,182,191,31,196,1099,5,172,10,179,83,21,20,21,186,177,195,193,188,199,62
,109,102};
        Array.Sort(d);
        int n = d.Length;

        double q1 = Med(d.Take(n / 2).ToArray());
        double q3 = Med(d.Skip((n + 1) / 2).ToArray());
        double iqr = q3 - q1;
        double low = q1 - 1.5 * iqr;
        double high = q3 + 1.5 * iqr;

        Console.WriteLine("Outliers:");
        foreach (var x in d)
        {
            if (x < low || x > high)
                Console.WriteLine(x);
        }
    }

    static double Med(int[] a) => a.Length % 2 == 0 ? (a[a.Length / 2] +
a[a.Length / 2 - 1]) / 2.0 : a[a.Length / 2];
}
```